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2014 IMSO Answers

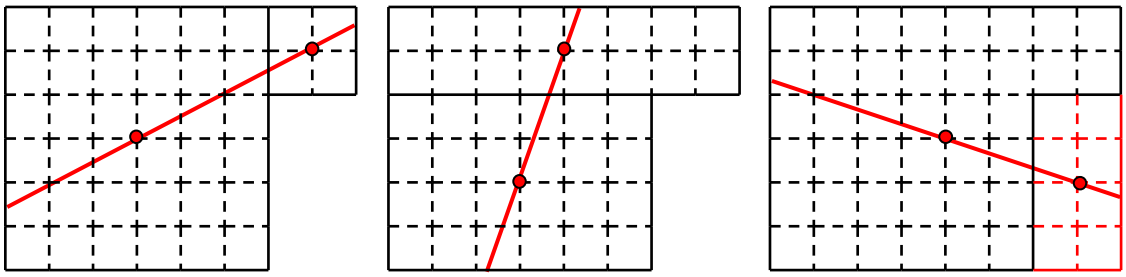
SHORT ANSWER

1.	1111111101	**2.**	$\frac{5}{9}$	**3.**	6 : 1	**4.**	68%	**5.**	17
6.	24	**7.**	242550	**8.**	402	**9.**	$\frac{3}{4}$	**10.**	258
11.	60, 16, 6, 6, 6 and 6	**12.**	15°	**13.**	65	**14.**	6750	**15.**	64
16.	62	**17.**	210	**18.**	158	**19.**	0	**20.**	2
21.	8.4	**22.**	384	**23.**	784913526	**24.**	96°	**25.**	2cm, 3cm, 5cm or 6cm

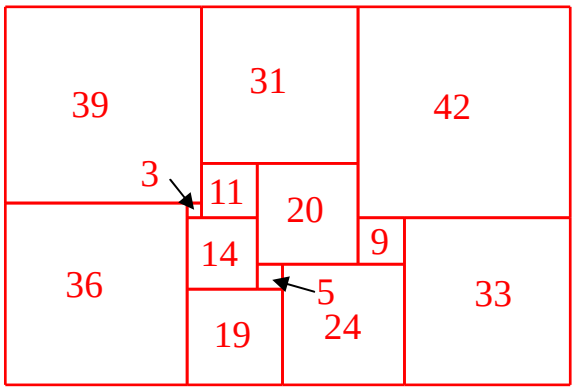
ESSAY PROBLEMS

1.	3000	**2.**	**7**	**3.**	0	**4.**	Holly works 3 hours and Molly works 2 hours		
5.	175	**6.**	40	**7.**	53	**8.**	$25\frac{620}{1001}$	**9.**	44°
10.	2012	**12.**	The maximum number of handshakes is 81 and the minimum number of handshakes is 17						

11.



13.



EXPLORATION PROBLEMS

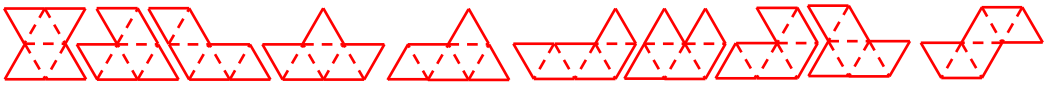
1.

2	2	2	2	2	2	12
2	2	2	2	2	1	11
2	2	2	2	1	0	9
2	1	1	0	0	0	4
1	1	0	0	0	0	2
1	0	0	0	0	0	1
10	8	7	6	5	3	

2.

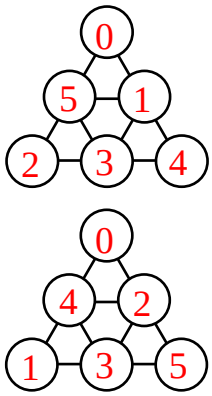
Denomination	1	7	49	343	2401	16807	117649	823543
Combination 1	1	1	3	3	3	3	1	1
Combination 2	8	0	3	3	3	3	1	1
Combination 3	1	8	2	3	3	3	1	1
Combination 4	8	7	2	3	3	3	1	1
Combination 5	1	1	3	3	3	3	8	0
Combination 6	8	0	3	3	3	3	8	0
Combination 7	1	8	2	3	3	3	8	0
Combination 8	8	7	2	3	3	3	8	0

3.

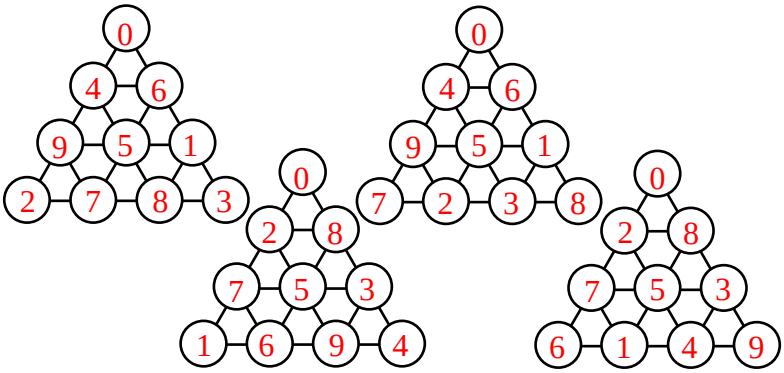


4. We have 17 such numbers, namely, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 22, 24, 26, 30, 36 and 60.

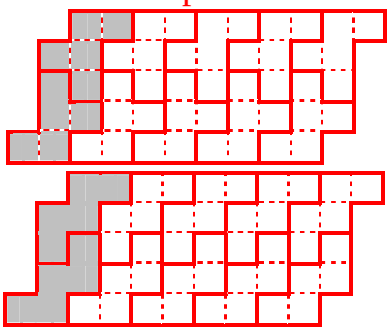
5. **(a)**



(b)



6. (a) The figure consists of the shaded parts in the two diagrams below.



(b) The figure constructed from ten copies of F is shown in the first diagram above. The figure constructed from ten copies of I is shown in the diagram below.

(c) The figure constructed from ten copies of W is shown in the second diagram above. The figure constructed from ten copies of I is shown in the diagram below.

